

1 **Habitat models of stream invertebrate community improve with predictors**
2 **of dam impacts and optimized gradient boosting machines**

Kei Nukazawa^{1,*}, Ryo Tanaka^{1,2}, Haruki Mineda¹

¹ Department of Civil and Environmental Engineering, Faculty of Engineering, University of

Miyazaki, Gakuen Kibanadai-nishi 1-1, Miyazaki 889-2192, Japan. (e-mail:

nukazawa.kei.b3@cc.miyazaki-u.ac.jp, haruki.mineda@gmail.com)

² West Japan Engineering Consultants, Inc., Watanabe-dori 1-1-1, Chuo, Fukuoka 810-0004,

Japan (e-mail: ryo-tanaka@wjec.co.jp)

*Corresponding author: Kei Nukazawa

Department of Civil and Environmental Engineering, Faculty of Engineering, University of

Miyazaki, Gakuen Kibanadai-nishi 1-1, Miyazaki 889-2192, Japan. (e-mail:

nukazawa.kei.b3@cc.miyazaki-u.ac.jp)

Abstract

Habitat models that can predict the distribution and habitat suitability of riverine organisms along environmental gradients are useful for watershed environmental management. However, dam impacts have not yet been considered in such models, despite the fact that dams have greatly altered important riverine habitats for organisms, including macroinvertebrate communities (e.g., riverbed materials and connectivity). This study developed catchment-scale habitat models of macroinvertebrate community with predictor variables that characterize dam impacts based on either simple geographical information system data (hereafter referred to as dam metrics) or physically simulated flow data using a distributed hydrological model. We studied the Omaru River catchment, where river flow has been significantly altered by multiple hydropower dams. Such an altered flow was represented by indicators of hydrologic alteration (IHA) using simulated flow data. We used multiple machine learning and regression techniques to model habitat distributions; random forest (RF), two gradient boosting techniques (XGBoost and LightGBM), a support vector machine, and logistic regression. The presence or absence of 170 macroinvertebrate taxa observed at 82 sites in different years (2018, 2019 and 2020) ($n = 107$) in the catchment was used as response variables. Among the modelling techniques, XGBoost and LightGBM with

optimized tree number parameters showed the highest mean accuracies among the taxa,
followed by RF. The accuracies of each algorithm greatly improved with the dam metrics and
IHA in both the community and habit groups, whereas the IHA tended to be more important
than the dam metrics based on the RF and LightGBM. This finding is useful for river
practitioners because they can consider environmental alterations in the decision-making
process for flood protection.

Keywords: Ecological niche modelling, LightGBM, Hydrological simulation, Indicators of
hydrologic alteration, Random forests, XGBoost

1. Introduction

Most riverine practices for environmental management have been implemented in limited local sections that are important for flood protection, and have not focused on the watershed scale. To expand this to broader contexts, habitat models (or species distribution models and ecological niche modelling) can provide useful information for decision-making. Habitat models are methods for estimating habitat suitability or the presence or absence of species at locations where biological surveys have not been conducted, by analysing the relationship between the spatial distribution of species and environmental attributes (e.g., temperature) based on statistical approaches (Elith and Leathwick, 2009). In the past, researchers have developed riverine habitat models targeting broad to catchment spatial scales (Bálint et al., 2011; Huang and Frimpong, 2016; Manel et al., 1999), while there are a few cases for those of relatively smaller scales (i.e., small to medium catchments). A primary reason for this bias is the lack of extensive data for statistical modelling such as limited biological data. For example, the Japanese Ministry of Land, Infrastructure, Transport and Tourism (MLIT) provides national census data for riverine organisms, including fish, benthos and plants (the River Environmental Database), while surveillance is performed in a few sections of limited rivers that are important for flood protection. Therefore, to develop well-trained habitat

models along broader environmental gradients at a basin scale, much sampling effort should be made to collect spatially extensive data from upstream to downstream.

The aforementioned problem of scarce data in riverine habitat models can be addressed by targeting macroinvertebrates because their sampling efforts are relatively low. In addition, riverine macroinvertebrates have limited dispersal ability compared to fish and are relatively sensitive to environmental alterations, suggesting that they have an advantage when structuring habitat models for assessing environmental impacts. In the past studies, there are some cases of constructing habitat models for macroinvertebrates (Cha et al., 2021; Domisch et al., 2013; Gogina and Zettler, 2010), though those at a catchment scale that is applicable to other regions (i.e., “transferable”, Hao et al., 2020) are yet to progress. This is partially due to insufficient consideration of environmental factors, such as hydrological factors and water temperature, which are more ecologically relevant to macroinvertebrate habitats (Guisan and Zimmermann, 2000; Jarnevich et al., 2015). While some earlier studies have applied such predictors (Kuemmerlen et al., 2015; Nukazawa et al., 2023, 2018), development of transferable models remains a challenge owing to their limited availability, .

Other important abiotic factors that preclude the transfer of riverine habitat models include a variety of anthropogenic influences such as water pollution and riverbed

degradation caused by land-use alteration and damming (Dudgeon et al., 2006; Jelks et al., 2008). The effects of dams are well-known as changes in stream environments through various pathways and the responses of organisms to these changes. Several previous studies have demonstrated the consequences of alterations, such as changes in macroinvertebrate communities through reduced algal production due to prolonged dam-induced turbid stream water (Armitage, 1977), and changes in macroinvertebrates and fish where streamflow is altered by dams (Nukazawa et al., 2020b). In reality, the interactive impacts of such factors affect natural environments and ecosystems; therefore, multiple impacts should be considered in a study framework, although such data are rarely acquired.

The present study applied previously developed simple dam metrics (Cooper et al., 2017) as well as indicators of hydrologic alteration (IHA) (Richter et al., 1996) to address the issue of acquiring data that represent complex dam impacts. Dam metrics can be derived from fundamental geographic information system (GIS) data and could represent combined impacts of a dam. For instance, metrics have successfully quantified the decline in stream connectivity caused by damming (Cote et al., 2009), the extent of flow regulation (Lehner et al., 2011) and the number of dams per river length in a catchment (hereafter referred to as dam density) (Van Looy et al., 2014). The IHA is an assemblage of 33 stream flow indices that assess given

flow alterations. It has been widely used in previous studies to assess the impacts of flow regulation on macroinvertebrates (Nukazawa et al., 2020b; Schneider and Petrin, 2017) and the influence of tributary confluences on fish population dynamics in the Missouri River (Pracheil et al., 2009). However, both metrics have been scarcely used in habitat modelling studies, specifically targeting multiple taxa with different ecologies in a biotic community, presumably because of less attention to such novel concepts (dam metrics) and less availability in site-specific flow data (IHA).

In this study, habitat models of macroinvertebrate community were developed using dam metrics and IHA in the Omaru River catchment in southwest Japan. We hypothesized that the contributions of the new predictors would vary among macroinvertebrate taxa owing to a variety of traits (e.g., habit) (Poff et al., 2006). Thus, we examined the extents to which the predictors contributed to the accuracy of the models in each habit group while optimizing a modelling structure for each taxon in partial modelling techniques (i.e., a parameter regarding tree number in gradient boosting machines). This study demonstrates the usefulness of such optimisation in modelling multiple taxa in a given community.

2. Materials and methods

2.1. Study area

We studied the Omaru River catchment in southwestern Japan (474 km²) (Fig. 1). The main stem stretches 75 km from the steep upland topology to the lower lands, with a relatively moderate slope (1/600 – 1/1000). The mean annual temperature and annual precipitation are approx. 17.6°C and 2,300 mm at the Takanabe meteorological station in the lowland (http://www.qsr.mlit.go.jp/miyazaki/kasen/omaru/gaiyou/omaru_saigai.html). The catchment population is approximately 31,000 and is mostly concentrated in the downstream area. The land use/land cover is dominated by forests (ca. 87%), followed by agricultural land (10%) and urbanised areas (3%). There are five dams in the main stem and tributary (Do River) that are used for flood control, hydropower generation and irrigation (Mineda et al., 2022).

2.2 Macroinvertebrate sampling

Macroinvertebrate sampling was performed in November 2018, 2019 and 2020. We fixed the season of sampling to exclude the effects of changes in community composition due to life-cycle events, such as emergence among campaigns, and to minimize the effects of disturbances caused by flooding that typically occur from summer to early autumn in this region. Indeed, during and before the sampling period, a negligible amount of rainfall was

only observed in 2018 (1.5 mm/day at upland Mikado meteorological station) and 2019 (4.5 mm/day at lowland Takanabe metrological station), and no rainfall was observed in 2020. We designed 82 sampling sites from the uppermost to downstream reaches, which were classified into 62 riffle sites (i.e., riffle dominated a site), 14 pool sites and 6 rifle-pool sites (i.e., riffle and pool dominated evenly) (Fig. 1). Of the 82 sites, we sampled twice at 25 sites (19 riffle sites and 6 pool sites) with broad environmental gradients in 2020 ($n = 107$).

We used a quadrat Surber sampler (25 cm x 25 cm, $\phi = 500 \mu\text{m}$) for macroinvertebrate sampling at riffles and pools. At the riffle sites, the sampler was set at four arbitrary points with flow velocity $> \text{ca. } 0.2 \text{ m/sec}$ and macroinvertebrates were sampled following the protocol in the River Environmental Database in MLIT, Japan. Briefly, gravel larger than 20 mm and cobbles were washed, and the riverbed was disrupted for 20 s while catching the benthos using an attached net. In the pool sites, the sampler was set at four arbitrary points with water depth $> \text{ca. } 0.35 \text{ m}$ and macroinvertebrates were sampled by disrupting the riverbed by foot for a minute. The sampled macroinvertebrates were identified to the lowest feasible taxonomic level (usually the species or genus level) using a microscope. Subsequently, taxa were classified into habit groups according to the literature (Table S1)

(Takemon, 2005), which is an optimized classification of Japanese macroinvertebrate taxa based on a previous study (Merritt and Cummins, 1996).

2.3 Environmental measurement

Environmental attributes were measured on the same dates as those used for macroinvertebrate sampling. We observed the current velocity (VR-30J, KENEK) and water depth at 4–5 points in each site and derived the means. The water temperature in each site was observed. We sampled 250 mL of flowing water and measured its pH and electrical conductivity (EC) (LAQUA, HORIBA).

In addition to the aforementioned measures, we visually determined the site categories in terms of riffles/pools, riverbank vegetation, canopy openness, levees, and bedrock. The riffle/pool consists of three classes; “riffle”, “pool” and “riffle-pool”. The riverbank vegetation consists of three classes; “no vegetation”, “passable” and “impassable”. The canopy openness consists of three classes; “no canopy”, “sunlight somewhat available” and “sunlight unavailable”. The levee consists of three classes; “no levee”, “more than half of the site section is covered by levee at either bank” and “more than half of the site section is

covered by levees at both banks”. The bedrock consists of two classes; “no bedrock” and “covered by bedrock”.

We measured the substrate material size at a riverbank sand bar as a proxy for the riverbed material size because of the considerable effort and cost of collecting riverbed data in our sampling design. For the measurement, we used a 1×1 m quadrat with 10 cm intervals; the major and minor axes of the materials under each cross section were measured, and the mean material size was derived.

2.4. GIS data

2.4.1. Fundamental geographical attributes

Land use was determined using an approximately 10-m resolution dataset of the global digital surface model observed by the Advanced Land Observing Satellite (ALOS) (https://www.eorc.jaxa.jp/ALOS/en/index_e.htm). We used hydrologically corrected elevation data at a spatial resolution of 1 s (approximately 30 m) (Yamazaki et al., 2020). Elevation was used to determine the catchment area and slope at a 250 m resolution throughout the Omaru River catchment. Here, we derived the two types of slopes; based on

the differences between the sampling site and either the grid cell with the highest altitude or the grid cell with the lowest altitude among the surrounding meshes.

2.4.2. Dam metrics

Sixteen dam metrics, developed in a previous study (Cooper et al., 2017) (Table S2), were applied to the Omaru River catchment. The metrics were derived using data on dam number and location, river network, reservoir storage and catchment area and classified into four groups. The first group included metrics for the sizes of river sections fragmented by dams (e.g., total segment length fragmented by dams, STOT). The second group included metrics for the number and density of dams (e.g., the number of dams upstream, UNCT). The third group included metrics for the distance to the dam (e.g., the distance to the nearest dam upstream of the main stem, UM2D). The fourth group included metrics for reservoir storage (e.g., total reservoir volume upstream per total length of upstream rivers, USR).

However, of these, seven metrics could not be computed at all sites because of their target locations (e.g., only computing for the main stem). Thus, to maintain the sample size for robust model development, these metrics were excluded from the habitat modelling procedures. River length from each site to up- and down-stream dams was measured using the

digital national land information for the river (nlftp.mlit.go.jp/ksj/) on QGIS ver.3.1.2 (QGIS Development Team, 2018). We acquired reservoir storage data from the webpages of the Kyusyu Electric Power Co. Inc. (http://www.kyuden.co.jp/company_outline_branch_miyazaki_initiative_power_01.html), and Miyazaki Prefecture (<https://www.pref.miyazaki.lg.jp/kasen/shakaikiban/kasen/dam.html>).

2.5. Flow metrics derived by hydrological simulations

As predictor variables, we used daily flow data at each sampling date, IHA, and change ratio of IHA. The IHA is classified into the five categories corresponding to impacts on riverine ecosystem: e.g., seasonal flow variation patterns (i.e., median flow at each month) and extreme flow (e.g., drought and flooding) magnitude and duration (e.g., maxima/minima of 1-90 day moving averages; 1-90-Max, 1-90-Min) (Richter et al., 1996). IHAs were computed for every calendar year corresponding to the sampling years (i.e., 2018, 2019 and 2020) (iha, R ver. 3.6.3) based on daily flow data simulated using a distributed hydrological model (Mineda et al., 2022). We considered the change ratio of each IHA with/without dams derived from hydrological analyses, whether using the initial conditions of dam-induced hydrological

alterations (e.g., dam outflow) or not. The change ratios were derived using the following

formulae:

(i) If the unit of IHA is m³/s, number, days,

$$R_1 = \frac{IHA_1 - IHA_0}{IHA_0} \times 100$$

(ii) If the unit of IHA is day,

$$R_2 = IHA_1 - IHA_0$$

where R_1 is the change ratio (%), R_2 is the extent of change (day), IHA_0 is the IHA derived

from the daily flow data computed without the dam data, and the IHA_1 is IHA derived from

the daily flow data computed with the dam data. Hereafter, R_1 and R_2 will be referred to as

C_IHA (Change ratio/extent of IHA). Partial C_IHA with missing values was excluded from

the subsequent modelling process. Detailed information on these metrics is presented in Table

S3.

2.6. Macroinvertebrate habitat modelling

2.6.1 Machine learning techniques

The flow of the analysis is shown in Fig. 2. We applied five regression and machine learning techniques to model the distributions of macroinvertebrates: random forest (RF), XGBoost (XGB), LightGBM (LGB), logistic regression (LGR) and a support vector machine (SVM).

The RF (randomForest, R ver. 4.1.2, R Core Team, 2021) (Breiman, 2001) is an ensemble learning technique with bootstrap aggregation that develops numerous decision trees by bootstrapping and predicts based on majority voting (classification) or averaging (regression). We adopted RF because it outperformed other machine learning classification techniques with a variety of response variables in many disciplines (Zhang et al., 2017), is suitable for modelling with many predictor variables (Feld et al., 2016), provides variable importance metrics and is generally robust to overfitting.

The XGB (xgboost, Python ver. 3.10.7) (Chen and Guestrin, 2016) is a variant of the gradient boosting machine (GBM), an ensemble machine learning model with a boosting technique. Boosting is an approach to sequentially develop decision trees while considering the loss functions of a previous tree to improve model performance. Although the fundamental GBM is generally time-consuming owing to its sequential modelling, XGB addresses this issue using weighted quantile sketch algorithm. In addition, as well as RF, it outperformed other machine learning classification techniques with a variety of response

variables in many disciplines (Zhang et al., 2017), and is suitable for modelling with many predictor variables (Maloney et al., 2012).

The LGB (lightgbm, Python ver. 3.10.7) (Ke et al., 2017) is based on GBM. LGB differs from XGB in generating processes of decision trees. While the XGB applies a “level-wise” procedure that splits all branches at a level and moves to the next level, the LGB applies a “leaf-wise” procedure that split nodes to minimize a loss, enabling efficient learning.

LGR (LogisticRegression, Python ver. 3.10.7) (Cox, 1958), also known as the generalized linear model, is a binary classification technique based on a sigmoid function.

SVM (SVC, Python ver. 3.10.7) (Cortes and Vapnik, 1995) is a classification method that maximizes the margins between classes. Both algorithms were conventionally used for binary classification problems prior to the proliferation of decision-tree-based models. Therefore, we adopted these algorithms for comparison with the decision-tree algorithms applied in this study.

2.6.2. Interpolating predictor variables

Before modelling the macroinvertebrate habitats, missing data in partial predictor variables were interpolated. Interpolation was performed for six environmental variables: flow velocity,

water depth, pH, EC, water temperature and substrate materials, using RF with 20 environmental attributes including GIS attributes. Because water temperature was measured at different times of the day, we added the sampling time of day as a predictor variable for water temperature. The accuracy of the models was evaluated based on ten times of random split cross-validation (80% training and 20% validation) in terms of linear mixed regressions between the observed and predicted values (Fig. S1). In general, the flow velocity, water depth, and substrate materials were relatively well modelled, whereas the accuracies of the others were limited.

2.6.3. Comparison of accuracies among the modelling algorithms

First, we built presence/absence habitat models for 171 macroinvertebrate taxa that were recorded as being present more than three samples with all predictor variables using the five modelling algorithms. Subsequently, the mean accuracies of the 171 taxa for each algorithm were compared based on ten times of random split cross-validation (80% training and 20% validation) in terms of the area under curve (AUC) of the receiver operating characteristic (ROCR, R ver. 4.1.2). The three algorithms with higher AUC values were used for downstream analyses.

2.6.4. Contributions of the dam metrics and IHA and parameter setting of machine learning

To evaluate the contributions of the dam metrics and IHA to predictions of macroinvertebrate habitats, the mean variable importance measures for each predictor variable among the 171 models were calculated based on RF and LGB (RF: MeanDecreaseAccuracy, LGB: split). Subsequently, we built 1) a model that excludes the dam index/IHA and half of the other variables with low importance (hereafter, Reference model), 2) a model that includes the dam metrics in the Reference model (hereafter, Dam model), 3) a model that includes the IHA in the Reference model (hereafter, IHA model) and 4) a model that includes both dam metrics and IHA in the Reference model (hereafter, Dam-IHA model), and compared the prediction accuracies of the four models. In this procedure, among the variables with a higher Spearman's rank correlation than the criterion (>0.7), variables with lower importance in RF and LGB were excluded to decrease model redundancy.

The hyperparameters of the machine-learning algorithms were tuned for the macroinvertebrate community. Although the sensitivity of the parameters in the RF was relatively low, we identified two important parameters: the number of features and the

number of trees as two and 500, respectively. For the XGB and LGB, sensitivity analyses were performed to identify the optimal hyperparameters. In both algorithms, owing to the substantial effort/computing cost of optimizing a vast number of changeable parameters (>100), we first focused on the parameters considered in previous studies (Bentéjac et al., 2021; Qiu et al., 2021) (Table S4). We varied one of the parameters individually, and the two most sensitive parameters were chosen for subsequent in-depth sensitivity analyses using both algorithms. In addition to the two sets of parameters, we used `early_stopping_round` (hereafter, ESR), a hyperparameter that stops the boosting process when it does not improve the performance, for the sensitivity analyses, as the ESR was found to greatly improve the accuracy of community model in our preliminary analyses. A hyperparameter for the number of trees (`num_boost_round`) was set to a large value (10,000) in this process so that the ESR works over a wide range of tree numbers. The accuracies were compared while changing the combination of the three parameters, and the combination with the best accuracy was identified as the final parameter set.

We used the same predictors in the sensitivity analyses as those in the dam-IHA model. The response variables for the sensitivity and contribution analyses of the dam metrics and IHA were presence or absence of 170 macroinvertebrate taxa. Note that a taxon that could

not be classified as our habit (*Chironomidae* sp.) was excluded during model development.

The accuracies of prediction among the 170 taxa were evaluated based on ten times of random split cross-validation (80% training and 20% validation) in terms of the AUC. To assess the contributions of the dam metrics and IHA, we used a paired t-test to compare the differences in the mean AUCs of the four models for each algorithm.

3. Results and Discussion

3.1. Comparison of prediction accuracy for each machine learning

The mean AUCs among the 171 macroinvertebrate taxa for each modelling algorithm with all predictor variables but without any parameter tuning exhibited that decision tree-based ensemble learning techniques (i.e., RF, XGB and LGB) outperformed the other algorithms with AUC values greater than 0.7, which is a criterion of fair prediction accuracy (Swets, 2015) (Fig. 3). The lower accuracies of the LGR and SVM were specifically characterized by taxa occurring in a few samples, suggesting the limitation of these algorithms in delineating classification boundaries. Thus, RF, XGB and LGB were used for downstream analyses.

3.2. Association between the environmental variables and macroinvertebrates

Among the predictors other than the dam metrics and IHA, altitude was the most important predictor variable based on the RF, followed by catchment area, current velocity and water depth (Fig. S2). The numbers of dams along the main stem downstream (DMCT) and per catchment area (UNDC) were important among the dam metrics. The median discharges in February (AF_Feb) and September (AF_Sep) were the most important IHA. Based on the LGB, the most important predictor variable, other than the dam metrics and IHA, was flow velocity, followed by water depth, altitude, and riverbed material. The catchment area of the river section fragmented by dams (SCA) and the UNDC were highly important dam metrics. The most important IHA metrics were the change ratios of rise rate for two consecutive days (C_Rise-rate) and the number of reversals in the flow rate for two consecutive days (C_Reversals).

These results highlight the importance of hydraulic variables, presumably because our sampling design targeted a broad range of hydraulic conditions (i.e., sampling in both riffles and pools). Hydraulics are critical habitat components for stream macroinvertebrates (Mérigoux and Dolédec, 2004; Poff et al., 2006); however, studies on riverine habitat modelling or species distribution modelling have rarely incorporated such metrics, whereas studies using model-based hydraulics and targeting fish exist (Nukazawa et al., 2023; Suzuki et al., 2021). This is probably because the typical sampling of macroinvertebrates was performed in flowing water.

These findings are potentially useful for expanding the conventional assessment framework to those also considering lentic water bodies.

In both RF and LGB, the IHA metrics tended to be more important than the dam metrics. This suggests that, among the influences of dams, the alteration of the flow regime could be a major influential factor for macroinvertebrates. While the IHA was more important in the RF, C_IHA tended to be more important in the LGB. This is probably due to the difference in the algorithms, specifically the difference in the methods of importance calculation. The IHA are inter-correlated and have similar characteristics in some metrics, whereas C_IHA are less inter-correlated. Therefore, with an importance method “split”, which counts the number of times a variable was used in model building, the relatively fair importance values among the C_IHA were produced in the LGB because a wide range of the C_IHA was used due to their different characteristics.

3.3. Selection of predictor variables used in the model

Redundant dam metrics and IHA with high Spearman's rank correlation coefficients were excluded considering the variable importance of RF and LGB. The correlation matrix among the metrics is shown in Fig. S3. Because the number of IHA metrics was large (>60), we

selected the two most important variables in RF and LGB for each IHA group. In brief, considering the cross-correlation and variable importance, the five variables such as elevation were selected in the Reference model, four additional variables such as UNDC were selected in the Dam model, nine additional variables such as Apr were selected in the IHA model, and both sets of variables in the Dam and IHA models were selected in the Dam-IHA model (Table S5).

3.4. Sensitivity analyses

XGB tended to show higher prediction accuracy, along with a relatively smaller subsample rate (SR), a proportion of randomly extracted samples (Fig. S4), suggesting that a smaller SR can avoid overfitting; thus better accuracy is obtained with the test data. The accuracy decreased when the SR changed from 0.5 to 0.15 (only 15% of the data used). This may be attributed to our relatively small sample size; the data for learning were insufficient and the accuracy decreased. Along with a higher learning rate (LR), the accuracy tended to increase, whereas the trend differed with $ESR = 0$ or a smaller SR. The accuracy improved as the ESR increased, regardless of the variations in SR and LR. In attempts at LGB, the accuracy increased with a smaller feature_fraction (FF), a ratio of randomly extracted features (i.e., predictors) (Fig. S5).

This implies that variables with high predictive ability are selectively used under a higher FF, resulting in a large model bias and low accuracy. Similar to XGB, higher LR and ESR generally increased accuracy.

The positive contribution of the higher LR found in both XGB and LGB was strongly associated with the number of trees and ESR. In general, if the LR is high, a smaller number of trees tends to show higher predictive performance. When ESR = 0, the models were developed with a designated number of trees (i.e., 10,000), which is a relatively large number; thus, the accuracy decreased with a higher LR. However, when the ESR was activated (ESR = 10–1,000), learning in many models stopped in the early learning stages (the number of trees was < 1,000). Therefore, the accuracy increased with a higher LR.

Based on the aforementioned considerations, the hyperparameters used in the contribution analyses of the dam metrics and IHA were identified, as listed in Table S4.

3.5. Contribution of the dam metrics and IHA to macroinvertebrate distributions

The mean AUC among 170 taxa of macroinvertebrates for all models, except for the Reference model of the RF, satisfied the criterion of good model accuracy of 0.7 (Fig. 5) (Swets, 2015).

The Dam-IHA model for LGB yielded the highest mean accuracy with an average AUC of

0.826. In the models with the dam metrics and IHA, all machine learning algorithms improved the average AUC. Thus, the dam metrics and IHA are useful for predicting macroinvertebrate community.

Figure 6 shows the mean AUC among the taxa of the eight habit groups for all models. In brief, the Reference models for RF, XGB and LGB attained the AUC criteria for the five, seven and eight groups, respectively. The accuracies for each algorithm were greatly improved with the dam metrics and IHA; the models with the dam metrics and IHA for RF, XGB and LGB attained the criterion for the 6–7, 7–8, and 8 groups, respectively. According to the results, the variable importance in modelling the habit groups for which the AUC significantly increased were subsequently evaluated.

The accuracy significantly increased with the IHA in each algorithm for Clingers_At ($P < 0.05$), a variant of Clingers with a sessile habit with/without a case, such as Simuliidae and *Limnocentropus* (Fig. 6). The important variables of the IHA were September median flow and base-flow index (Fig. 7). The September flow reflects a typhoon-induced high flow; therefore, the extent of shear stress in September may constrain the settlement of the group. In addition, a lower base-flow index is probably unsuitable for this group since extremely low flow diminishes higher current area, which is a preferable habitat for Clingers_At.

By incorporating the dam metrics and IHA, the accuracies for Crawlers and Clingers_Gl tended to increase significantly ($P < 0.05$). These results suggest that many taxa in these habit groups determine their distribution in response to dam-induced alterations, including flow regime alterations. The one-day maximum flow rate tended to be important for Crawlers. Crawlers such as Ephemerellidae crawl on the surfaces of larger substrate materials. This finding implies that riverbed disturbances by extremely high flows, which promote their passive drifts, also regenerate material structures and affect their distributions. The rise-rate for two consecutive days tended to be important for Clingers_Gl, a group that includes Heptageniidae and Psephenidae. This metric decreases downstream of the dams because the sections receive only residual flow, except during the flooding period (Mineda et al., 2022). Thus, this manipulation suppressed pulsed floods, resulting in a decrease in the metric, and consequently affecting the distributions of Clingers_Gl. A previous study in an adjacent catchment reported that taxa belonging to Clingers_Gl significantly decreased downstream of the dams due to habitat degradation associated with algal growth (Nukazawa et al., 2020a). Hence, such suppressed flow promoted algal growth and constrained the inhabitation of the group.

4. Conclusions

The present study demonstrates that the metrics of dam-induced impacts are useful for habitat modelling or ecological niche analyses of stream macroinvertebrate community at a catchment scale. To do so, we compared cutting-edge ensemble learning techniques (random forests, XGBoost and LightGBM) and found that the gradient boosting methods with a tuned parameter for optimising tree numbers to each taxon were promising in predicting the distributions of biotic community. The habitat models developed are likely to be more applicable to other catchments or periods (i.e., transferable) (Hao et al., 2020) than conventional habitat models that typically used geographical attributes such as altitude and latitude because a vast number of small to large catchments were altered by dams and weirs in the Anthropocene. Furthermore, the higher predictive capability of the dam metrics and flow regime is specifically useful to river practitioners because they will be able to consider environmental alterations in flood protection decision-making. To this end, future studies should test the transferability of the models in terms of time and space.

431 **Acknowledgements**

432 This work was partially supported by the Ministry of Education, Science, Sports and Culture
433 through a Grant-in-Aid for Scientific Research (grant no. 20H00256).

434

435

436

References

- Armitage, P.D., 1977. Invertebrate drift in the regulated River Tees, and an unregulated tributary Maize Beck, below Cow Green dam. *Freshw. Biol.* 7, 167–183.
- Bálint, M., Domisch, S., Engelhardt, C.H.M., Haase, P., Lehrian, S., Sauer, J., Theissinger, K., Pauls, S.U., Nowak, C., 2011. Cryptic biodiversity loss linked to global climate change. *Nat. Clim. Chang.* 1, 313–318. <https://doi.org/10.1038/NCLIMATE1191>
- Bentéjac, C., Csörgő, A., Martínez-Muñoz, G., 2021. A comparative analysis of gradient boosting algorithms, *Artificial Intelligence Review*. Springer Netherlands. <https://doi.org/10.1007/s10462-020-09896-5>
- Breiman, L., 2001. Random forests. *Mach. Learn.* 5–32. <https://doi.org/10.1023/A:1010933404324>
- Cha, Y.K., Shin, J., Go, B.G., Lee, D.S., Kim, Y.W., Kim, T.H., Park, Y.S., 2021. An interpretable machine learning method for supporting ecosystem management: Application to species distribution models of freshwater macroinvertebrates. *J. Environ. Manage.* 291, 112719. <https://doi.org/10.1016/j.jenvman.2021.112719>
- Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system. *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.* 13-17-Aug, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Cooper, A.R., Infante, D.M., Daniel, W.M., Wehrly, K.E., Wang, L., Brenden, T.O., 2017. Assessment of dam effects on streams and fish assemblages of the conterminous USA. *Sci. Total Environ.* 586, 879–889. <https://doi.org/10.1016/j.scitotenv.2017.02.067>
- Cortes, C., Vapnik, V., 1995. Support-vector networks. *Mach. Learn.* 20, 273–297. <https://doi.org/10.1088/1742-6596/628/1/012073>
- Cote, D., Kehler, D.G., Bourne, C., Wiersma, Y.F., 2009. A new measure of longitudinal connectivity for stream networks. *Landsc. Ecol.* 24, 101–113. <https://doi.org/10.1007/s10980-008-9283-y>
- Cox, D.R., 1958. The regression analysis of binary sequences. *J. R. Stat. Soc. Ser. B* 20, 215–242.
- Domisch, S., Kuemmerlen, M., Jähnig, S.C., Haase, P., 2013. Choice of study area and predictors affect habitat suitability projections, but not the performance of species distribution models of stream biota. *Ecol. Modell.* 257, 1–10. <https://doi.org/10.1016/j.ecolmodel.2013.02.019>
- Dudgeon, D., Arthington, A.H., Gessner, M.O., Kawabata, Z.-I., Knowler, D.J., Lévêque, C.,

- Naiman, R.J., Prieur-Richard, A.-H., Soto, D., Stiassny, M.L.J., Sullivan, C.A., 2006. Freshwater biodiversity: importance, threats, status and conservation challenges. *Biol. Rev. Camb. Philos. Soc.* 81, 163–182. <https://doi.org/10.1017/S1464793105006950>
- Elith, J., Leathwick, J.R., 2009. Species Distribution Models: Ecological Explanation and Prediction Across Space and Time. *Annu. Rev. Ecol. Evol. Syst.* 40, 677–697. <https://doi.org/10.1146/annurev.ecolsys.110308.120159>
- Feld, C.K., Segurado, P., Gutiérrez-Cánovas, C., 2016. Analysing the impact of multiple stressors in aquatic biomonitoring data: A ‘cookbook’ with applications in R. *Sci. Total Environ.* 573, 1320–1339. <https://doi.org/10.1016/j.scitotenv.2016.06.243>
- Gogina, M., Zettler, M.L., 2010. Diversity and distribution of benthic macrofauna in the Baltic Sea. Data inventory and its use for species distribution modelling and prediction. *J. Sea Res.* 64, 313–321. <https://doi.org/10.1016/j.seares.2010.04.005>
- Guisan, A., Zimmermann, N.E., 2000. Predictive habitat distribution models in ecology. *Ecol. Modell.* 135, 147–186. [https://doi.org/10.1016/S0304-3800\(00\)00354-9](https://doi.org/10.1016/S0304-3800(00)00354-9)
- Hao, T., Elith, J., Lahoz-Monfort, J.J., Guillera-Aroita, G., 2020. Testing whether ensemble modelling is advantageous for maximising predictive performance of species distribution models. *Ecography (Cop.)*. 43, 549–558. <https://doi.org/10.1111/ecog.04890>
- Huang, J., Frimpong, E.A., 2016. Limited transferability of stream-fish distribution models among river catchments: Reasons and implications. *Freshw. Biol.* 61, 729–744. <https://doi.org/10.1111/fwb.12743>
- Jarnevich, C.S., Stohlgren, T.J., Kumar, S., Morissette, J.T., Holcombe, T.R., 2015. Caveats for correlative species distribution modeling. *Ecol. Inform.* 29, 6–15. <https://doi.org/10.1016/j.ecoinf.2015.06.007>
- Jelks, H.L., Walsh, S.J., Burkhead, N.M., Contreras-Balderas, S., Diaz-Pardo, E., Hendrickson, D.A., Lyons, J., Mandrak, N.E., McCormick, F., Nelson, J.S., Platania, S.P., Porter, B.A., Renaud, C.B., Schmitter-Soto, J.J., Taylor, E.B., Warren, M.L., 2008. Conservation status of imperiled North American freshwater and diadromous fishes. *Fisheries* 33, 372–407. <https://doi.org/10.1577/1548-8446-33.8.372>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* 2017-Decem, 3147–3155.
- Kuemmerlen, M., Schmalz, B., Cai, Q., Haase, P., 2015. An attack on two fronts : predicting how changes in land use and climate affect the distribution of stream macroinvertebrates. *Freshw. Biol.* 60, 1443–1458. <https://doi.org/10.1111/fwb.12580>

- Lehner, B., Liermann, C.R., Revenga, C., Vörösmarty, C., Fekete, B., Crouzet, P., Döll, P., Endejan, M., Frenken, K., Magome, J., Nilsson, C., Robertson, J.C., Rödel, R., Sindorf, N., Wissler, D., 2011. High-resolution mapping of the world's reservoirs and dams for sustainable river-flow management. *Front. Ecol. Environ.* 9, 494–502. <https://doi.org/10.1890/100125>
- Maloney, K.O., Schmid, M., Weller, D.E., 2012. Applying additive modelling and gradient boosting to assess the effects of watershed and reach characteristics on riverine assemblages. *Methods Ecol. Evol.* 3, 116–128. <https://doi.org/10.1111/j.2041-210X.2011.00124.x>
- Manel, S., Dias, J.M., Ormerod, S.J., 1999. Comparing discriminant analysis, neural networks and logistic regression for predicting species distributions: A case study with a Himalayan river bird. *Ecol. Modell.* 120, 337–347. [https://doi.org/10.1016/S0304-3800\(99\)00113-1](https://doi.org/10.1016/S0304-3800(99)00113-1)
- Mérigoux, S., Dolédec, S., 2004. Hydraulic requirements of stream communities : a case study on invertebrates. *Freshw. Biol.* 49, 600–613.
- Merritt, R.W., Cummins, K.W., 1996. An Introduction to the Aquatic Insects of North America. 3rd ed., in: Kendall Hunt. <https://doi.org/10.2307/4053>
- Mineda, H., Nukazawa, K., Suzuki, Y., 2022. Assessing watershed-scale environmental flow alterations induced by dams and climate change using a distributed hydrological model. *Authorea Prepr.*
- Nukazawa, K., Arai, R., Kazama, S., Takemon, Y., 2018. Projection of invertebrate populations in the headwater streams of a temperate catchment under a changing climate. *Sci. Total Environ.* 642, 610–618. <https://doi.org/10.1016/j.scitotenv.2018.06.109>
- Nukazawa, K., Chiu, M.-C., Kazama, S., Watanabe, K., 2023. Contrasting adaptive genetic consequences of stream insects under changing climate. *Sci. Total Environ.* 872, 162258. <https://doi.org/10.1016/j.scitotenv.2023.162258>
- Nukazawa, K., Kajiwar, S., Saito, T., Suzuki, Y., 2020a. Preliminary assessment of the impacts of sediment sluicing events on stream insects in the Mimi River, Japan. *Ecol. Eng.* 145, 105726. <https://doi.org/10.1016/j.ecoleng.2020.105726>
- Nukazawa, K., Shirasaka, K., Kajiwar, S., Saito, T., Irie, M., Suzuki, Y., 2020b. Gradients of flow regulation shape community structures of stream fishes and insects within a catchment subject to typhoon events. *Sci. Total Environ.* 748, 141398. <https://doi.org/10.1016/j.scitotenv.2020.141398>
- Poff, N.L., Olden, J.D., Vieira, N.K.M., Finn, D.S., Simmons, M.P., Kondratieff, B.C., 2006.

- Functional trait niches of North American lotic insects: Traits-based ecological applications in light of phylogenetic relationships. *J. North Am. Benthol. Soc.* 25, 730–755. [https://doi.org/10.1899/0887-3593\(2006\)025\[0730:ftnona\]2.0.co;2](https://doi.org/10.1899/0887-3593(2006)025[0730:ftnona]2.0.co;2)
- Pracheil, B.M., Pegg, M.A., Mestl, G.E., 2009. Tributaries influence recruitment of fish in large rivers. *Ecol. Freshw. Fish* 18, 603–609. <https://doi.org/10.1111/j.1600-0633.2009.00376.x>
- Qiu, Y., Ding, S., Yao, N., Gu, D., Li, X., 2021. HFS-LightGBM: A machine learning model based on hybrid feature selection for classifying ICU patient readmissions. *Expert Syst.* 38. <https://doi.org/10.1111/exsy.12658>
- Richter, B.D., Baumgartner, J. V., Powell, J., Braun, D.P., 1996. A method for assessing hydrologic alteration within ecosystems. *Conserv. Biol.* 10, 1163–1174. <https://doi.org/doi.org/10.1046/j.1523-1739.1996.10041163.x>
- Schneider, S.C., Petrin, Z., 2017. Effects of flow regime on benthic algae and macroinvertebrates - A comparison between regulated and unregulated rivers. *Sci. Total Environ.* 579, 1059–1072. <https://doi.org/10.1016/j.scitotenv.2016.11.060>
- Suzuki, K., Ishiyama, N., Koizumi, I., Nakamura, F., 2021. Combined effects of summer water temperature and current velocity on the distribution of a cold-water-adapted sculpin (*cottus nozawae*). *Water (Switzerland)* 13, 1–11. <https://doi.org/10.3390/w13070975>
- Swets, J.A., 2015. Measuring the Accuracy of Diagnostic Systems. *Science (80-.)*. 240, 1285–1293.
- Takemon, Y., 2005. Life-type concept and functional feeding groups of benthos communities as indicators of lotic ecosystem conditions. *Japanese J. Ecol.* 55, 189–197.
- Van Looy, K., Tormos, T., Souchon, Y., 2014. Disentangling dam impacts in river networks. *Ecol. Indic.* 37, 10–20. <https://doi.org/10.1016/j.ecolind.2013.10.006>
- Yamazaki, D., Togashi, S., Takeshima, A., Sayama, T., 2020. High-resolution flow direction map of Japan. *J. Japan Soc. Civ. Eng.* 8, 234–240. https://doi.org/10.2208/JOURNALOFJSCE.8.1_234
- Zhang, C., Liu, C., Zhang, X., Almpandis, G., 2017. An up-to-date comparison of state-of-the-art classification algorithms. *Expert Syst. Appl.* 82, 128–150. <https://doi.org/10.1016/j.eswa.2017.04.003>

Figures

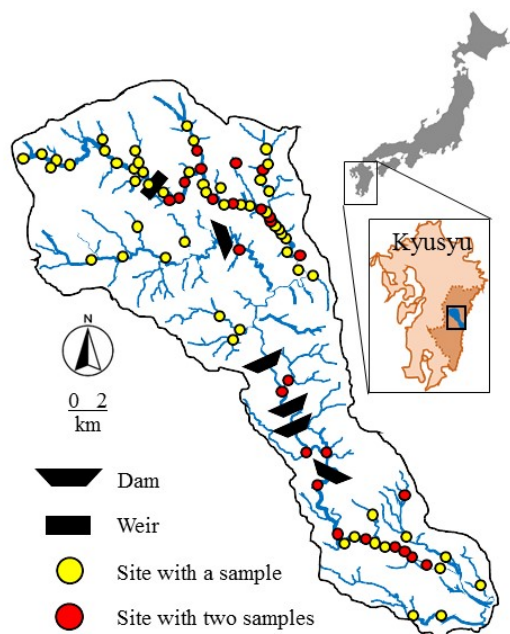


Figure 1 Study area

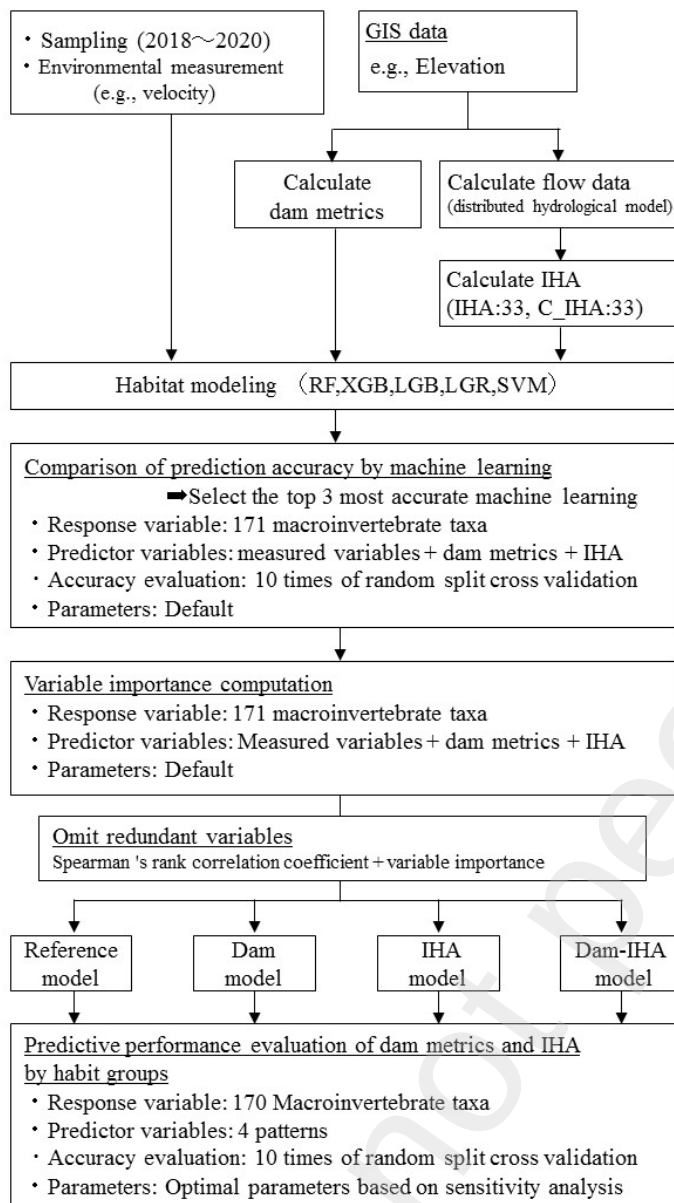


Figure 2 Study flow

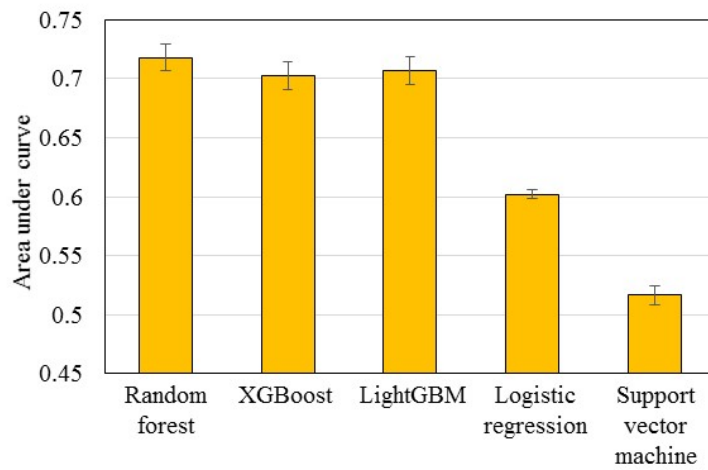


Figure 3 Mean AUC among models of 171 macroinvertebrate taxa with all predictors for each algorithm

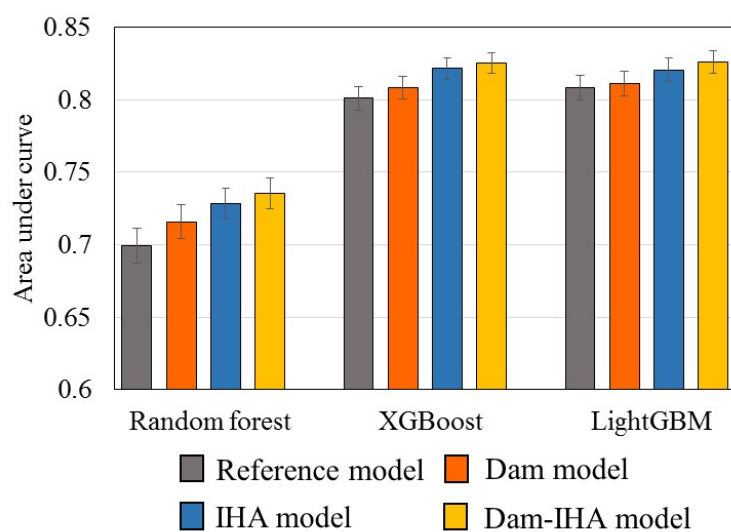


Figure 4 Mean area under curve (AUC) among models of 170 macroinvertebrate taxa with selected predictors for the random forest, XGBoost and LightGBM.

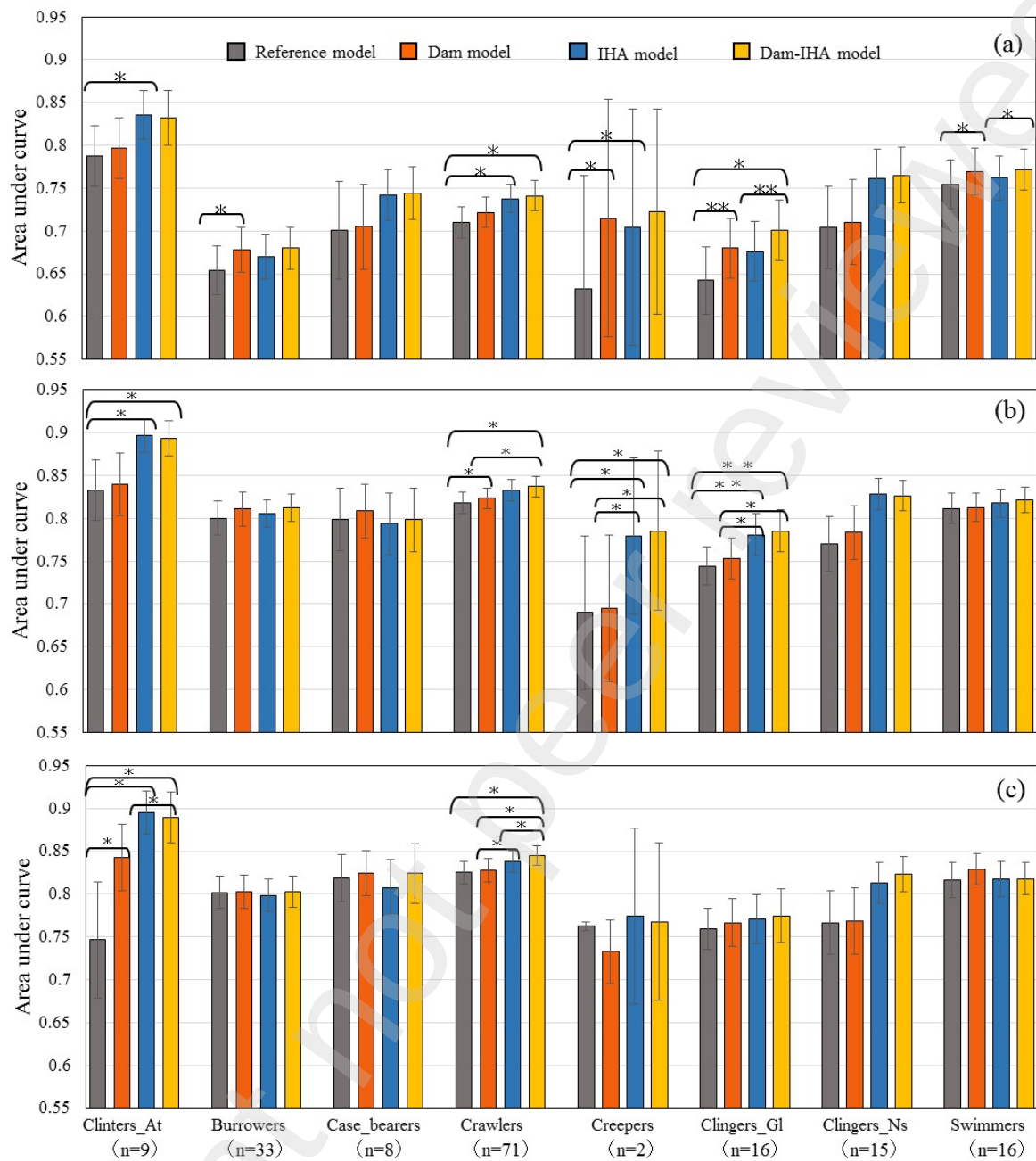


Figure 5 Mean area under curve (AUC) among models of eight habit groups with selected predictors for the random forest (a), XGBoost (b) and LightGBM (c). Within each group, significant difference between the models is highlighted ($p < 0.05$: *, $p < 0.01$: **).

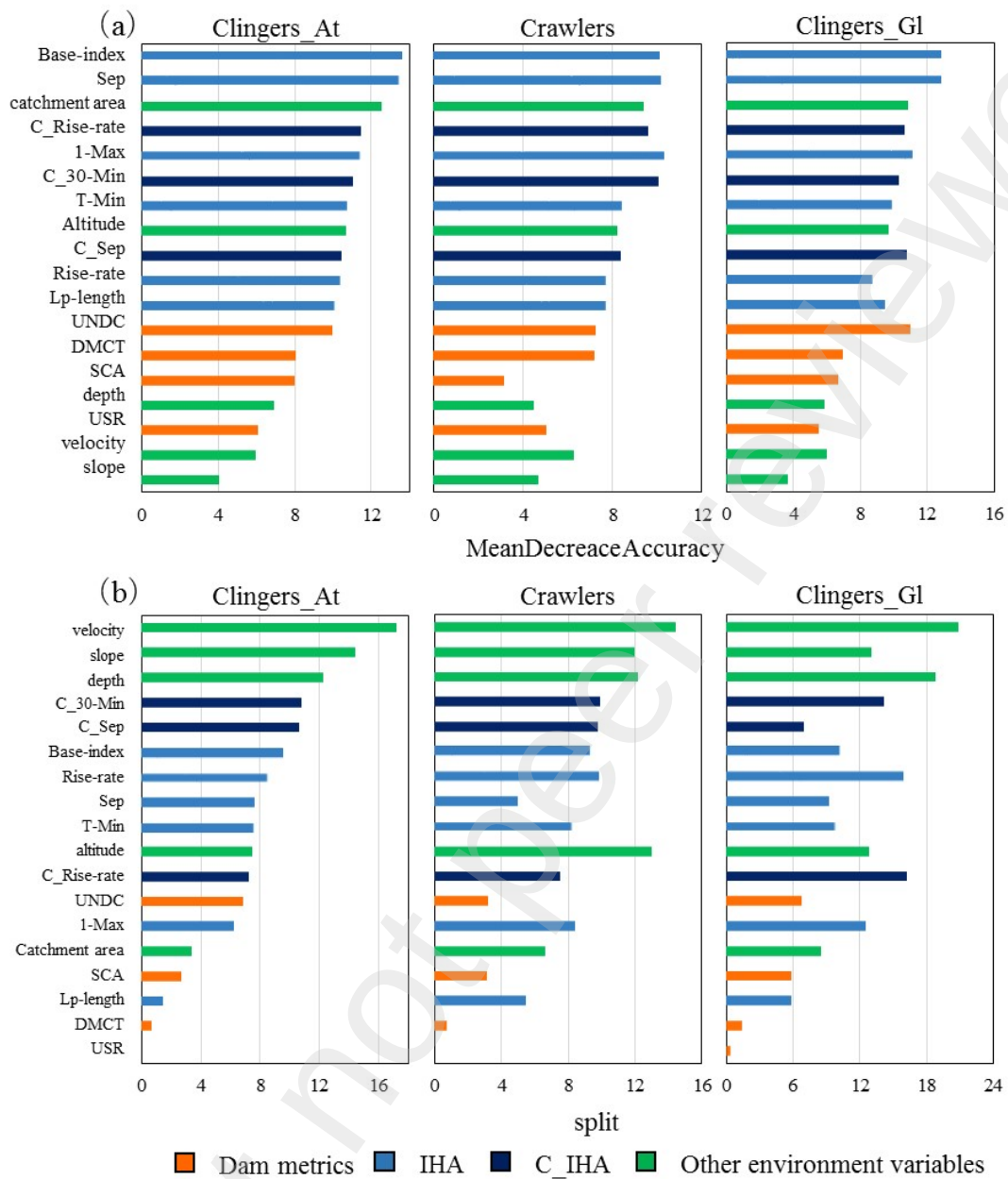


Figure 6 Mean variable importance among models of three habit groups for random forest (a) and LightGBM (b). Here range of negative importance value is excluded for visibility reason.